



STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

Page No :A-1 of 2
IS : 2062



CM/L-0057534
MTL-II/QR-2

TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL STEEL

Sold-to-Party:
BM BSO JAIPUR
S-16A LANDMARK 4TH FLOOR OPP.GUJARATI SAMAJ MAHAVIR
JAIPUR 302001

T.C.No : RCL/MTL/PLM/80931377
T.C. Date : 24.03.2026
SALES ORDER NO : 1100584216
DA No : **80931377**

24

Ship-to-Party:

WHM JAIPUR,Jaipur.

Wagon No / Trailer No : ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V. NOTCH (L) Impact Values (J)							
mm	mm	mm			330 MPa Min	490 MPa	Min22(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
20	2500	12000	3517590/2	413215	410	545	23	--	88	90	102	93	RT	OK	2.0 T	
				413215	422	546	24	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517575/2	413225	430	555	24	--	90	98	86	91	RT	OK	2.0 T	
				413225	425	546	24	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517582/1	413225	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3517598/1	413228	422	542	24	--	94	88	90	91	RT	OK	2.0 T	
				413228	423	544	23	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517601/1	413228	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3518653/1	413331	447	566	25	--	116	110	118	115	RT	OK	2.0 T	
				413331	432	568	23	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3518655/2	413331	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3518659/1	413331	--	--	--	--	--	--	--	--	--	--	--	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
413215	0.17	0.020	0.023	1.40	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.046	< 0.005	65	< 0.0005	0.41
413225	0.17	0.018	0.020	1.37	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	64	< 0.0005	0.41
413228	0.17	0.020	0.019	1.41	0.25	0.023	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.044	< 0.005	69	< 0.0005	0.41
413331	0.17	0.024	0.018	1.40	0.28	0.033	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	66	< 0.0005	0.41

Total Pieces Page / Sales Order : 8 / 8 Total Pieces / TC : 12

The Material supplied conforms to the standard rolling and weight tolerances.,

Validity unknown

Digitally signed by SSAJU
Date: 2026.03.24 18:56:19 IST
Reason: Authorized Signatory RCL, BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

Page No :A-2 of 2

IS : 2062



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MTL-II/QR-2

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Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No

: RCL/MTL/PLM/80931377

T.C. Date

: 24.03.2026

SALES ORDER NO

: 1100584216

DA No

: 80931377

Wagon No / Trailer No

: ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Size of impact test specimen: 10x10x55 mm(above 10mm thickness) ; 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

Validity unknown

Digitally signed by SAILJU
Date: 2026.03.24 18:56:19 IST
Reason: Authorized Signatory RCL,
BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

Page No : B-1 of 1

IS : 2062



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MTL-II/QR-2

**TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL**

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP: GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR, Jaipur.

T.C.No

: RCL/MTL/PLM/80931377

T.C. Date

: 24.03.2026

SALES ORDER NO

: 1100584217

DA No

: **80931377**

Wagon No / Trailer No

: ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			330 MPa Min	490 MPa	Min22(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
25	2500	12000	3517780/1	413670	447	575	23	--	98	90	88	92	RT	OK	2.0 T	
				413670	434	560	24	--	--	--	--	--	--	OK	2.0 T	
25	2500	12000	3517790/1	413670	--	--	--	--	--	--	--	--	--	OK	2.0 T	
25	2500	12000	3517876/1	413845	408	529	24	--	92	96	88	92	RT	OK	2.0 T	
				413845	422	549	23	--	--	--	--	--	--	OK	2.0 T	
25	2500	12000	3517888/1	413848	416	534	23	--	100	104	102	102	RT	OK	2.0 T	
				413848	400	527	24	--	--	--	--	--	--	OK	2.0 T	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
413670	0.17	0.015	0.024	1.48	0.30	0.020	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.044	< 0.005	68	< 0.0005	0.43
413845	0.15	0.024	0.022	1.38	0.26	0.028	< 0.02	< 0.015	< 0.015	< 0.015	0.034	< 0.010	< 0.005	70	< 0.0005	0.38
413848	0.17	0.030	0.025	1.35	0.22	0.033	< 0.02	< 0.015	< 0.015	< 0.015	0.030	< 0.010	< 0.005	67	< 0.0005	0.40
Total Pieces Page / Sales Order : 1 / 1																

Total Pieces Page / Sales Order : 4 / 4

Total Pieces / TC

: 12

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Size of impact test specimen: 10x10x55 mm (above 10mm thickness); 7.5x10x55 mm (upto 10mm thickness),

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